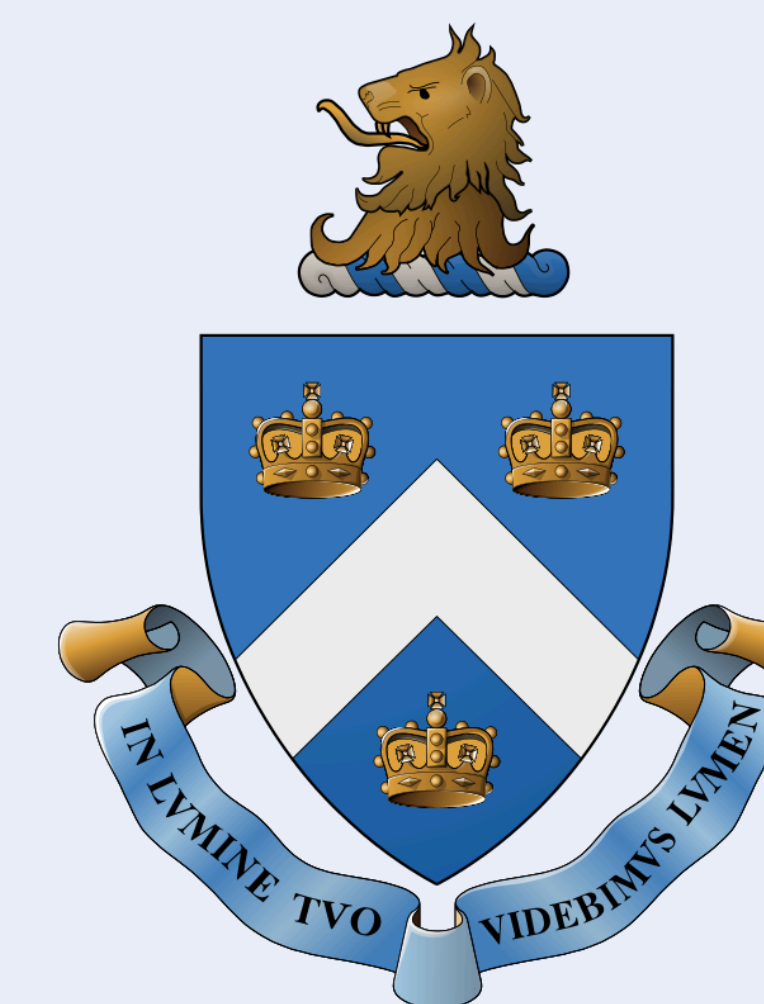


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SAGE: A Top-Down Bottom-Up Knowledge-Grounded User Simulator for Multi-turn Agent Evaluation

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Problem

Evaluating multi-turn interactive agents demands costly human assessment. Existing user simulators model **generic users** and overlook the domain-specific business knowledge needed to reproduce realistic behavior — yielding interactions of limited evaluation utility.

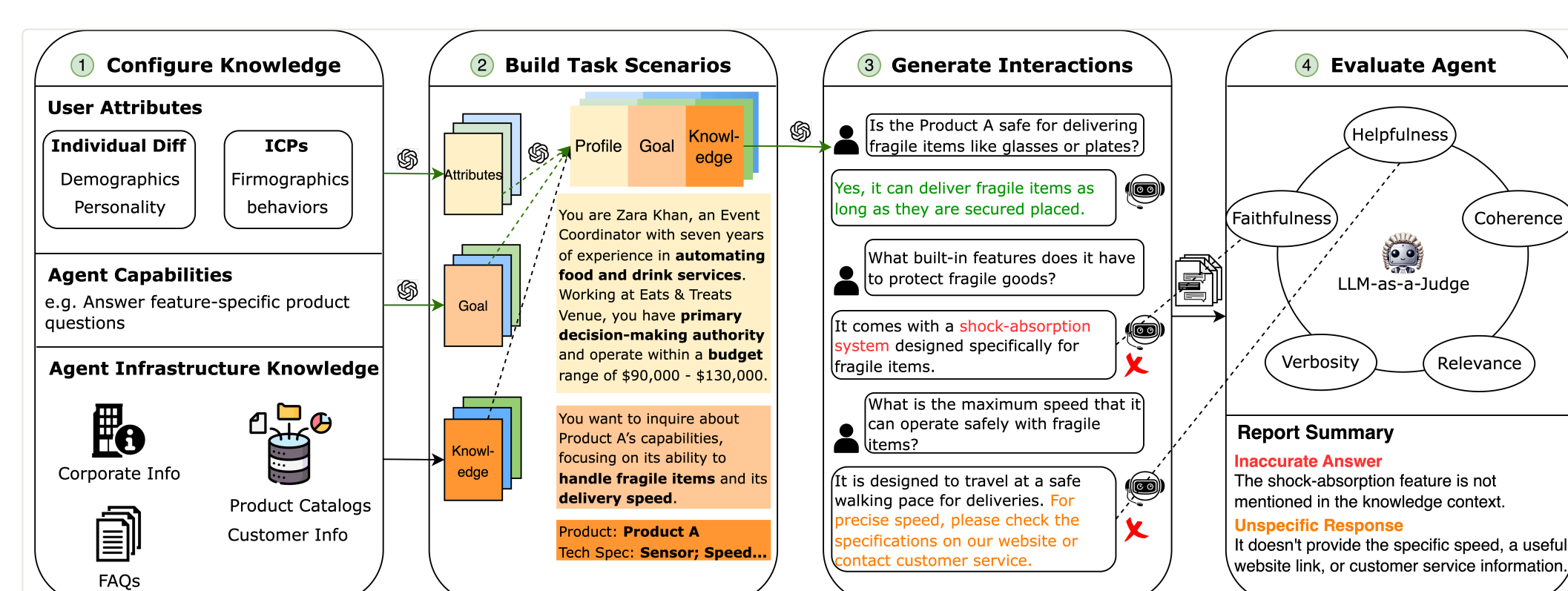
Evaluation is the bottleneck — human testing does not scale.

Motivation

- Real agents are deployed in **specialized business settings** where behavior is shaped by *who* the user is (role, demographics, budget) and *what* company-specific information they seek.
- Simulators that ignore business design principles — ideal customer profiles, product knowledge — produce interactions that are less realistic and hard to use for rigorous evaluation.

Key idea: inject **top-down** customer profiles *and* **bottom-up** agent infrastructure so the simulated user behaves like a real customer of *this* business.

Method



SAGE overview: configure knowledge → build scenarios → generate interactions → evaluate agent.

1 Configure Knowledge *top-down & bottom-up*

- User attributes:** Big-Five personality & demographics (individual differences).
- ICPs:** firmographics, decision authority, budget — the most valuable customers.
- Agent infrastructure:** the company's own knowledge, given to the simulator.

Catalogs FAQs **Personality** Budget

2 Build Task Scenarios

goal → knowledge → profile

- Pick a user goal** from agent capabilities.
- Select** the most relevant infrastructure knowledge (hierarchical, clustering-based, *diverse-yet-relevant*).
- Build** a coherent user profile, then **compose** a natural-language scenario prompt.

3 Generate Interactions *shared environment*

Simulator and agent converse in a shared environment over the interaction history and infrastructure knowledge, ending on a **###STOP###** sentinel or turn budget. An LLM-as-a-Judge scores each reply on five dimensions:

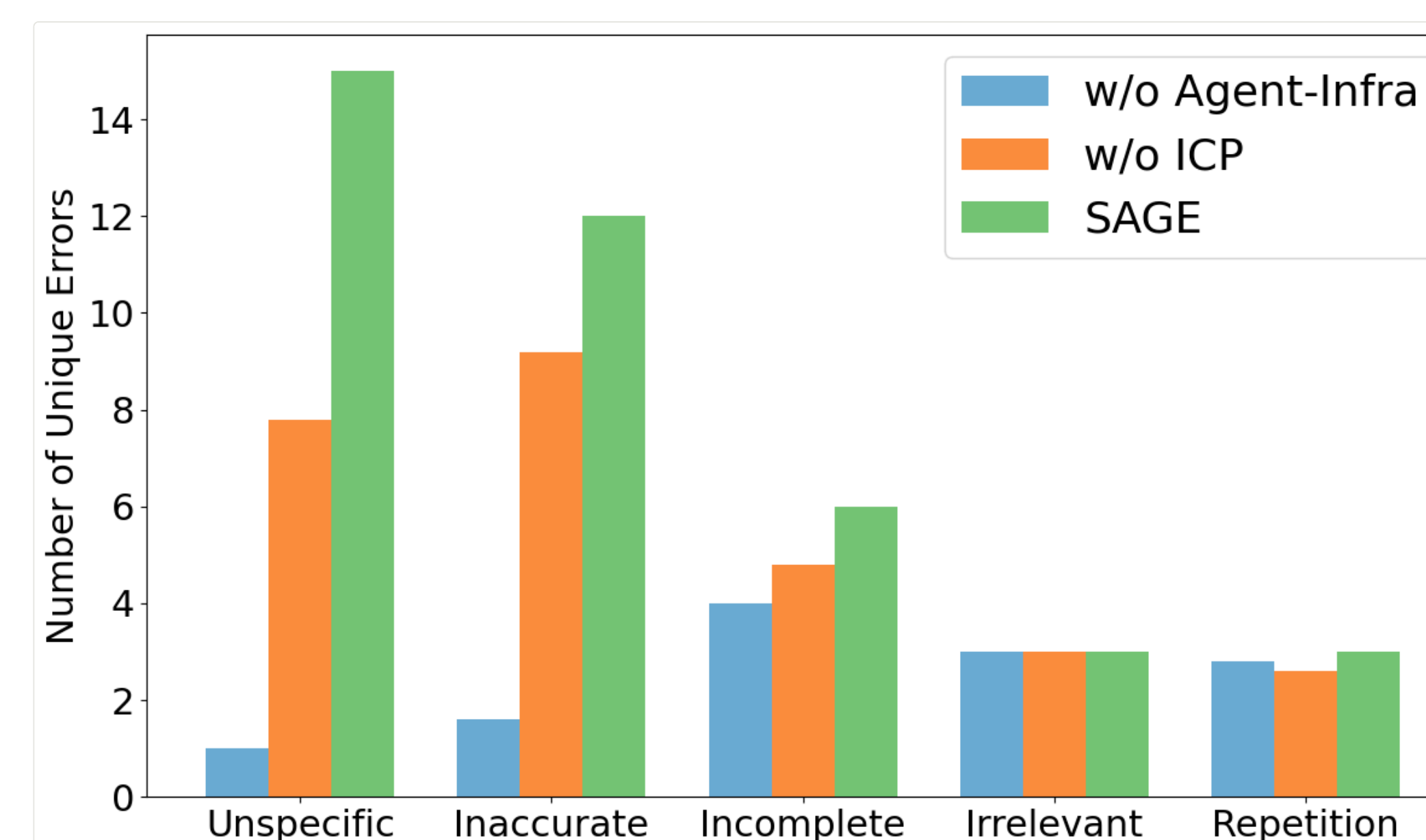
$$s(t) \in \{0, \dots, 4\} \text{ keep } t \text{ as a bug iff } s(t) < 2$$

scored on helpfulness, coherence, verbosity, relevance, faithfulness

4 Evaluate & Identify Bugs *LLM-as-a-Judge*

- Low-scoring turns are summarized into **fine-grained, unique bug statements** with linked turns.
- LLM bug lists match human experts at **F1 0.73** (Fleiss' κ 0.85).

94% of flagged low-scoring turns were real bugs (human-verified)



Unique errors by category (RAG sales agent).

2.58

human-likeness
(vs 2.90)

86.0

MTLD
(human 86.8)

150

interactions
/ condition

Key Results

METHOD	RAG BUGS	SHOP BUGS
w/o Agent-Infra	13.6	13.0
w/o ICP	29.0	14.4
SAGE	38.6	18.0

Headline Numbers

+33%
more agent
bugs identified

vs. existing user-simulation methods

38.6 bugs
(RAG)

2.58 human-like

0.73 bug-ID
F1

Takeaway

Grounding a user simulator in real business knowledge — customer profiles plus agent infrastructure — makes interactions more realistic and finds more agent bugs than generic simulators or even human testers.

Business-grounded simulation > generic simulation for bug-finding

Scan to Read

Paper



Code

